

Milind Agrawal

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Surat, Gujarat

SUMMARY

Software developer building full-stack applications (MERN) and researching quantum machine learning for biometric verification. Delivered 3 end-to-end web applications and a QkSVM pipeline benchmarked on LFW at NIT Surat. Seeking SDE internship roles at product-focused teams.

EDUCATION

SRM Institute of Science and Technology

B.Tech in Electronics and Communication Engineering

CGPA: 8.72

Chennai, Tamil Nadu

[2027]

EXPERIENCE

NIT, Surat

[05/2026 - 06/2026]

Research Intern

- Designed and evaluated a Quantum Kernel Support Vector Machine (QkSVM) pipeline for biometric face verification using 512-D ArcFace embeddings, characterizing a peak verification accuracy of 93.74% against a near-ceiling classical baseline of 99.18% on the Labeled Faces in the Wild (LFW) dataset.
- Architected a reproducible PennyLane simulation pipeline for 8 and 12-qubit feature maps, demonstrating a 4x scaling improvement in True Accept Rate (TAR jumped from 0.14 to 0.57 at a strict False Accept Rate of 10^{-3}) by scaling the qubit and training-pair budget.
- Technologies Used:** Python, PennyLane, InsightFace (ArcFace), Scikit-Learn, NumPy, Principal Component Analysis (PCA)

PROJECTS

AI-POWERED JOB PORTAL (FULL STACK + GENAI)

[Link]

- Engineered a full-stack recruitment platform with distinct candidate, recruiter, and admin modules, implementing secure authentication, job postings, application tracking, and resume uploads
- Integrated an AI-powered interview preparation feature into the platform using Google Gemini, achieving consistent 20-25s report generation regardless of resume length, and validated reliability through repeated-run consistency testing while identifying API rate-limit constraints under sustained use.
- Technologies Used:** React, Redux Toolkit, Node.js, Express, MongoDB, @google/genai, Cloudinary

REAL-TIME CHAT APPLICATION

[Link]

- Engineered a highly responsive full-stack messaging platform utilizing the MERN stack and Socket.IO to enable instantaneous, bidirectional communication and live online user presence tracking.
- Hardened application security by implementing JWT authentication delivered via secure, httpOnly cookies, and managed complex cross-origin resource sharing (CORS) using Vercel proxy rewrite configurations.
- Technologies Used:** React (Vite), Node.js, Express, MongoDB, Socket.IO, JWT, Cloudinary, Multer

REAL-TIME COLLABORATIVE CODE EDITOR (CRDT-BASED)

[Link]

- Built a real-time collaborative code editor enabling multiple users to simultaneously edit the same document with automatic conflict resolution, integrating Yjs (CRDT) with Monaco Editor and Socket.IO for operation-based synchronization across clients.
- Verified seamless, imperceptible-latency sync across 10 concurrent editors with zero data loss on overlapping edits; containerized the application and deployed it via AWS ECR and ECS.
- Technologies Used:** React, Vite, TailwindCSS, Monaco Editor, Yjs, Node.js, Express, Socket.IO, Docker, AWS (ECR, ECS)

SKILLS

- Languages:** Python, JavaScript, SQL
- Frameworks And Technologies:** React, Express, Node.js, MongoDB, Amazon Web Services
- Tools:** Git/Github, Docker, Postman
- Soft Skills:** Leadership, Requirements Gathering, Adaptability, Quick Learning

ACHIEVEMENTS/CERTIFICATES

- 1st Place Winner**, University Hackathon - Organized by the Department of Physics, SRM University

[04/2024]

- The Complete Python Bootcamp From Zero to Hero in Python

[Link]